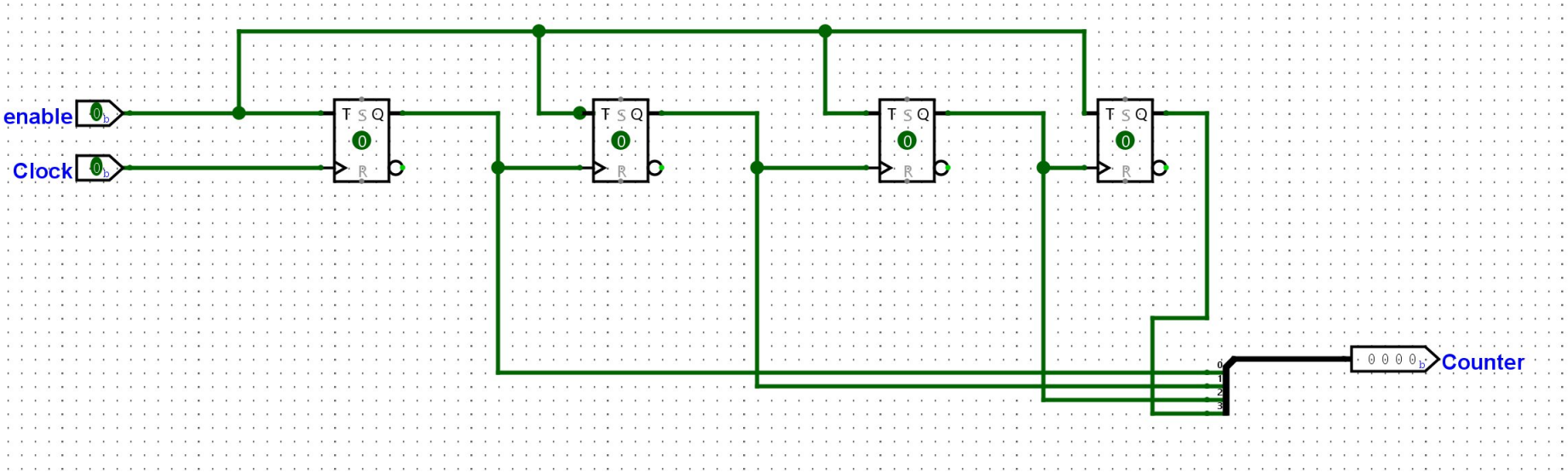
Lab 6

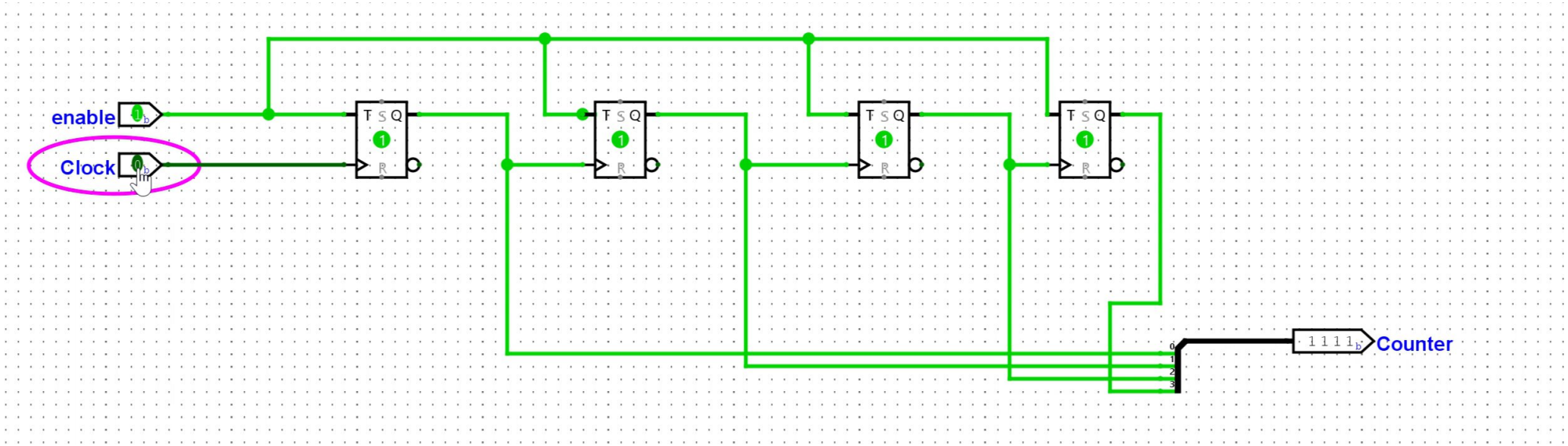
Andrew Walker

Down Counter from 15

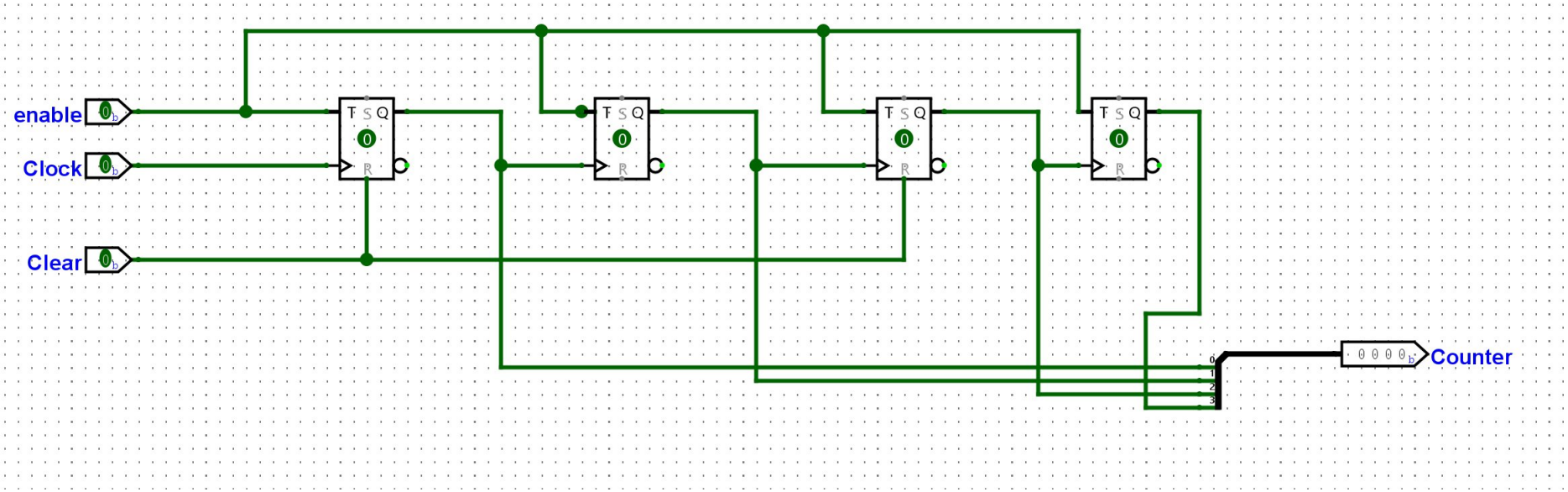


This is a 4-bit down counter that starts from 15 and ends displaying zero. After displaying zero it resets to 15 in binary.

Down Counter Video

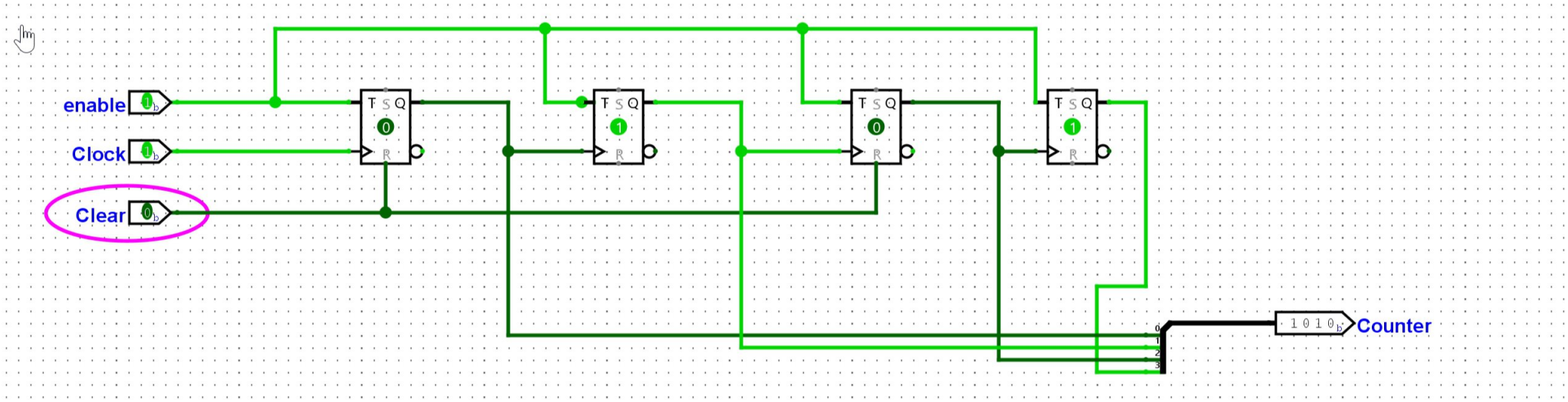


Down Counter with Clear

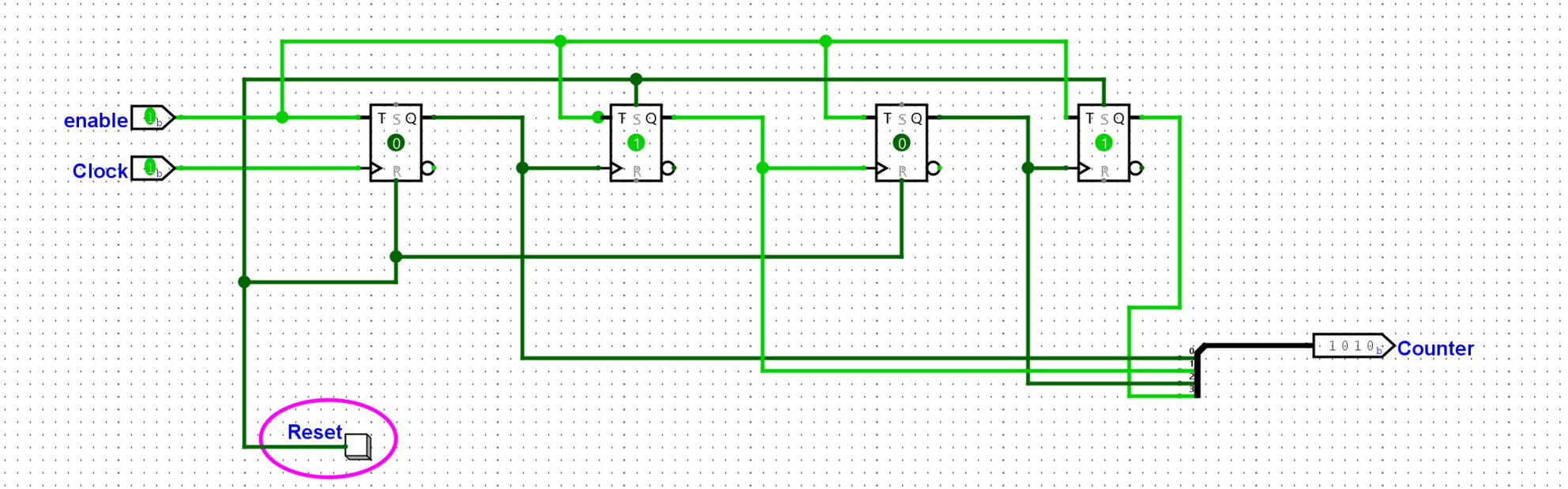


Same down counter but I have added a clear function. This clear function only resets to ten if the number is greater than ten not less. This happens because I only have the reset pins connected to the clear without the preset pins connected.

Down Counter with Clear Video

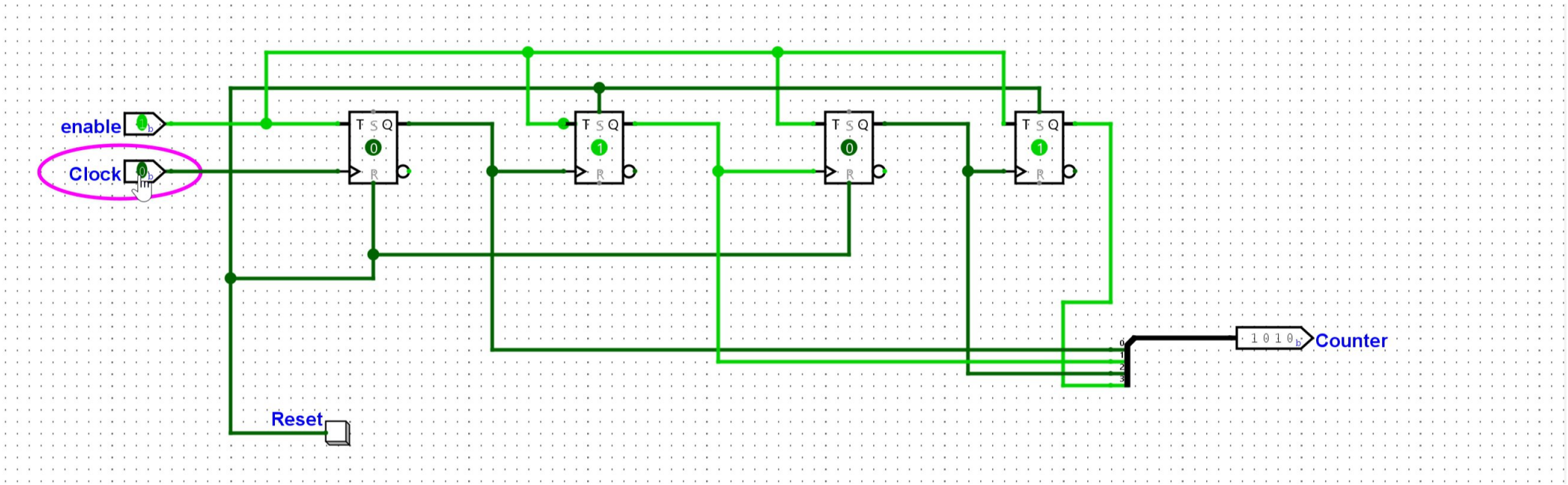


Button Reset

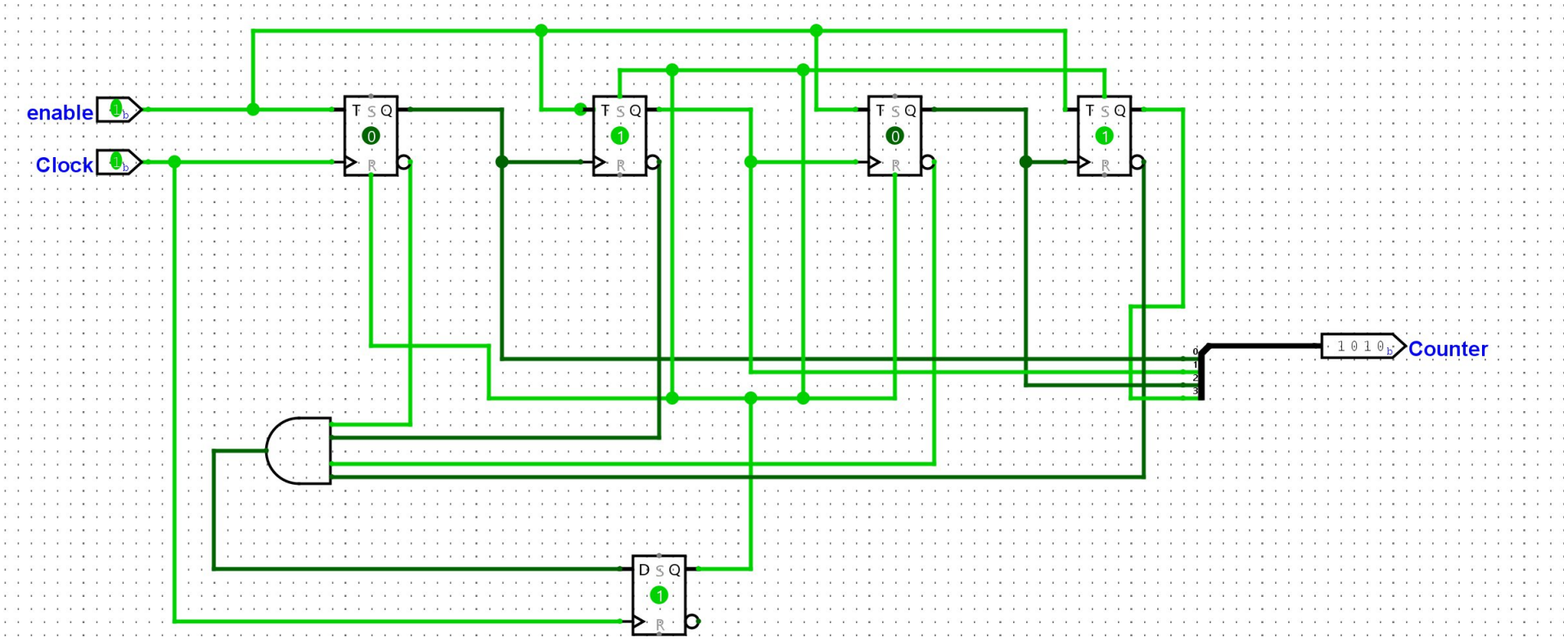


The button acts like the clear except it happens no matter what number it is at. I fixed this by connecting the preset pins. Doing this tells the circuit that when the button is pressed, it should reset to ten.

Button Reset Video

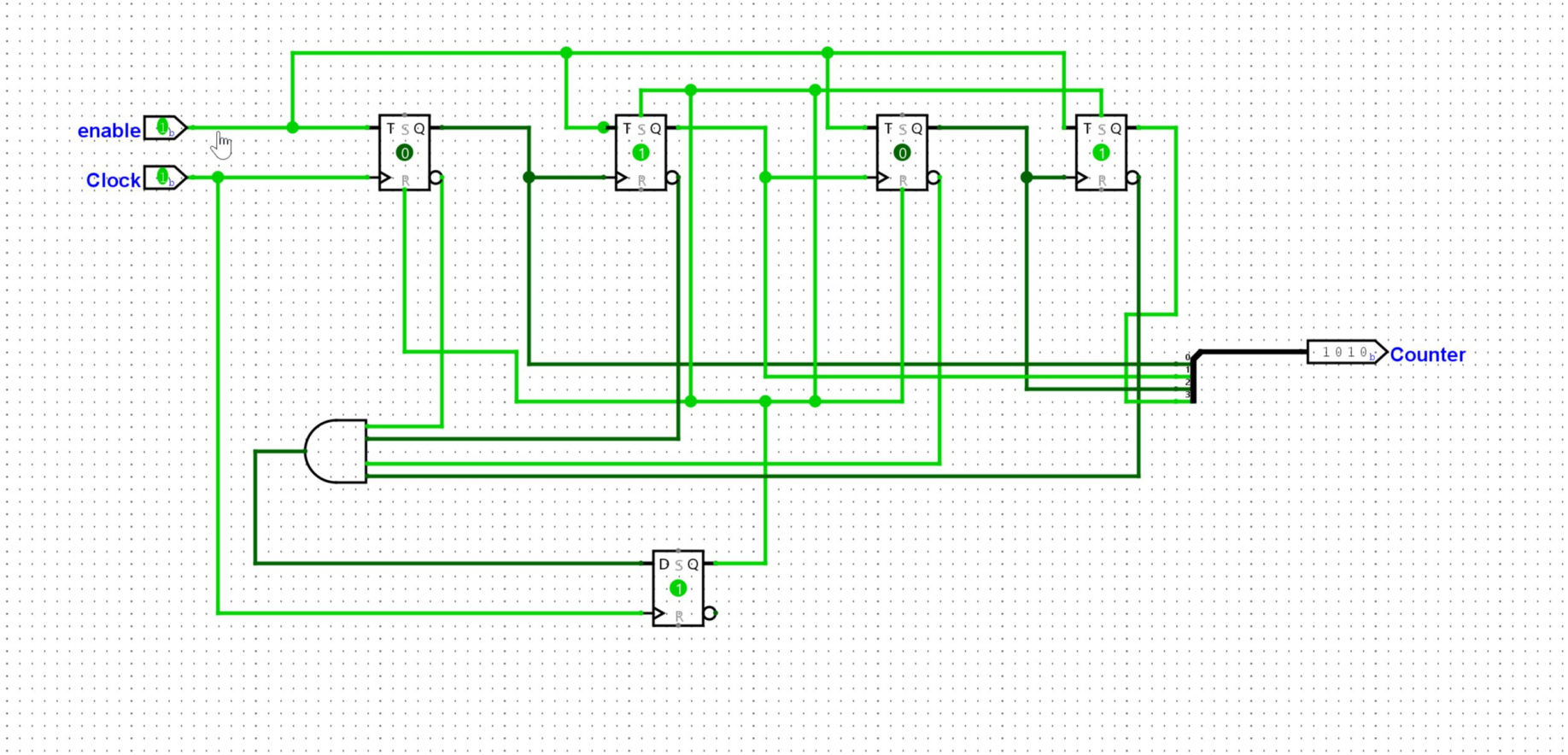


Automatic Reset



This is my final asynchronous down counter with an automatic reset. I added two memory functions with the AND gate and the D flip flop. The AND gate tells the circuit to reset when the counter hits zero. Without the D flip flop, the counter would never display zero. The D flip flop was added to tell the circuit to wait on more period and then reset.

Automatic Reset Video



Full Circuit Counter

